Contents

1. How it works
2. Installing Plex Media Server
3. Installing Plex Downloader
4. Two devices: Plex on one, Downloader on another
5. Everyday use
6. Updating, repairing and diagnostics
7. Easier SSH login (keys)
8. Access from outside your home (VPN)
9. Security in your home network

Plex Downloader – guide

This guide takes you step by step from a clean Raspberry Pi to a working Plex library with a download manager. Replace the example addresses and names (`192.168.1.10`, `pi`, `SSD 1TB`) with your own.

---

1. How it works

Plex Media Server plays movies and TV shows on your TV, phone and computer. **Plex Downloader** is a panel in the browser that:

- downloads files from direct links (several connections at once, resumes after a restart),
- names them straight away the way Plex expects (`Title (Year)/Title (Year).mkv`, `Show/Season 01/Show - S01E02.mkv`),
- has a **File manager**: moves between disks with SHA-256 verification, renames, changes posters, deletes.

Three ways to set it up

Setup	For whom	Notes
External disks (recommended)	Plex and Downloader on one Raspberry Pi, library on an SSD/HDD or USB stick	fastest and simplest
SD card (not recommended)	a quick test without disks	the card is small, slow and wears out with large writes
Two devices	Plex on one computer, Downloader on another (e.g. a Pi Zero 2 W)	library shared over SMB; downloads are slower

You need: a Raspberry Pi with Raspberry Pi OS (32 or 64-bit), SSH access and an internet connection. Use the official power supply – a weak power supply is the most common reason for a Pi that freezes.

2. Installing Plex Media Server

If Plex is already running, go to chapter 3.

The easy way: fix.sh

After installing Plex Downloader (chapter 3) run `./fix.sh` → **7 Additional modules** → **Plex Media Server** → **Install**. The script adds the official Plex repository and installs the server. At the end it shows `http://<pi-address>:32400/web` – open it in the browser and sign in with your Plex account.

By hand

```
sudo apt install -y curl gnupg
curl -fsSL https://downloads.plex.tv/plex-keys/PlexSign.key | sudo gpg --dearmor -o
/usr/share/keyrings/plex.gpg
echo "deb [signed-by=/usr/share/keyrings/plex.gpg] https://downloads.plex.tv/repo/deb
public main" | sudo tee /etc/apt/sources.list.d/plexmediaserver.list
sudo apt update && sudo apt install -y plexmediaserver
```

External disk: permanent mount

The disk has to be **mounted permanently** (also after a restart), otherwise Plex and Downloader will not find the library.

1. Find the disk name and UUID:

   ```
   lsblk -o NAME,SIZE,FSTYPE,LABEL,UUID,MOUNTPOINT
   ```

2. Create a mount point, e.g. `/mnt/ssd1`:

   ```
   sudo mkdir -p /mnt/ssd1
   ```

3. Add a line to `/etc/fstab` (`sudo nano /etc/fstab`) with the UUID from step 1:

   ```
   UUID=1234-ABCD  /mnt/ssd1  ext4  defaults,nofail  0  2
   ```

   For a disk formatted as exFAT or NTFS write `exfat` or `ntfs3` instead of `ext4` and add `uid=pi,gid=pi`, e.g. `defaults,nofail,uid=pi,gid=pi`.

4. Mount it and check – no error means it will also be mounted after a restart:

   ```
   sudo mount -a && df -h /mnt/ssd1
   ```

5. Create the library folders (or let the wizard do it – chapter 3):

```
mkdir -p "/mnt/ssd1/Plex/Movies" "/mnt/ssd1/Plex/TV Shows" "/mnt/ssd1/Plex/Kids"
```

Permissions: Plex runs as the user `plex`, Plex Downloader as you (e.g. `pi`). Plex must be able to read the folders, Downloader to write:

```
sudo chown -R pi:pi /mnt/ssd1/Plex
sudo chmod -R 755 /mnt/ssd1/Plex
```

In the Plex app add the libraries: **Movies** → `/mnt/ssd1/Plex/Movies`, **TV Shows** → `/mnt/ssd1/Plex/TV Shows` and so on.

Test setup: library on the SD card

Not recommended for everyday use. The SD card is small, slower than a disk and wears out faster. If it fills up completely, the whole system may stop working – always keep a few GB free.

The library then lives in your home folder, e.g. `/home/pi/Plex/Movies`. The Plex Downloader wizard has a ready option **"Library on the SD card"**, and Settings → Disks has a **"+ SD card (not recommended)"** button. To copy files from your computer, share the `Plex` folder over SMB (chapter 4).

3. Installing Plex Downloader

Log in to the Raspberry Pi over SSH (e.g. `ssh pi@192.168.1.10`, details in chapter 7) and run:

```
cd ~
git clone https://github.com/pwmeess/plex-downloader.git
cd plex-downloader
chmod +x setup.sh update.sh fix.sh
./setup.sh
```

The script:

1. asks for the language (1 – English, 2 – Polski),
2. installs the system packages (aria2, Python, git) and ffmpeg – **on a Pi Zero this can take 5–15 minutes**,
3. creates the Python environment; on weak Wi-Fi `Retrying…` warnings are normal – if the download fails, run `./setup.sh` again, nothing is lost,
4. asks for a **username and password** for the panel (min. 6 characters),
5. starts the service and shows the panel address, e.g. **http://192.168.1.10:8800**.

> **The port is 8800**, not 8080.

Setup wizard

The wizard opens after the first login. Run it again any time with **Settings → Account → Run setup wizard** – it then starts from scratch (only the categories and the saved TMDB key are kept). **Close without saving** leaves the current settings unchanged.

1. **Language and theme.** The first step also has **“I have a settings file – load it”** – handy for a second Raspberry Pi or after a reinstall (chapter 5). Without a file the wizard guides you through everything.
2. **Where the library is:** external disks, SD card or Plex on another device.
3. **Disks and folders**
    - The wizard checks the mounted disks (or network shares) and **ticks** the ones with library folders by itself. It reads only folder names – never their contents.
    - It recognises common names and variants: `Movies`, `Filmy`, `Cinema`, `TV Shows`, `Series`, `Kids`, `Home Video` = `Home Videos` … Music and photo folders are skipped.
    - Next to each category: ⚙ renames the category (e.g. “Kids” → “Concerts”), ✕ means “not on this disk”, ↺ brings it back with a suggested folder, **+ Add category** creates a new one.
    - Unusual folders next to the library (e.g. `Movies02`) are shown as **“+ Movies02”** buttons – a click adds them as a new category.
    - **Create missing folders when finishing** is off by default. Folders that do not exist and will not be created are left out.
    - **Next** only needs a Movies or TV Shows folder on any disk. **Skip – set up disks later** is for when there is no library yet.
4. **Plex server.** Servers in your home network are found automatically. Add a server by hand only when it is on another subnet or behind a VPN.
5. **TMDB key** (optional, free): create an account at themoviedb.org → Settings → API → create a key (“Personal use” is enough) and paste the **API Key** or the **API Read Access Token**. The wizard checks it right away: ✓ **Key OK** / ✕ **Invalid key**. With a key you get the movie preview card and posters in the File manager.
    - **Generate posters myself from video frames (needs ffmpeg)** – for home videos and titles TMDB does not know. Not recommended right away: about 10–15 s per movie on a Pi 4 / Pi 400, longer on a Pi Zero. A poster from a frame for a single movie can always be made in the Edit window (chapter 5).
6. **Summary → Finish.** The previous settings are kept as a restore point.

---

4. Two devices: Plex on one, Downloader on another

Example: the **Plex Pi** (`192.168.1.10`) has the disks, the **Downloader Pi** (`192.168.1.20`, e.g. a Pi Zero 2 W) writes to them over the network.

1. **Fixed IP addresses.** Reserve fixed addresses for both devices in your router (DHCP reservation).
2. **Share the library on the Plex Pi** (Samba).
3. **Mount the share on the Downloader Pi** (cifs, permanently in fstab).
4. Run the wizard and choose **"Plex on another device"**.

> Downloads are slower in this setup: every byte travels over the network twice. Over 2.4 GHz Wi-Fi (Pi Zero 2 W) expect a few MB/s.

Sharing folders (Samba) – on the Plex Pi

`./fix.sh` → **7 Additional modules** → **Samba** → **Install**, or `sudo apt install -y samba`. Then add a share at the end of `/etc/samba/smb.conf` (`sudo nano /etc/samba/smb.conf`):

```
[Plex SSD1]
    path = /mnt/ssd1/Plex
    valid users = pi
    read only = no
    force user = pi
```

Set the Samba password for the user and restart the service:

```
sudo smbpasswd -a pi
sudo systemctl restart smbd
```

`valid users` lets only this user in – never share the library with everyone.

Mounting the share – on the Downloader Pi

```
sudo apt install -y cifs-utils
sudo mkdir -p /mnt/ssd1
nano ~/.smbcredentials
```

Put two lines into `~/.smbcredentials` and save (Ctrl+O, Enter, Ctrl+X):

```
username=pi
password=YOUR_SAMBA_PASSWORD
```

```
chmod 600 ~/.smbcredentials
sudo mount -t cifs "//192.168.1.10/Plex SSD1" /mnt/ssd1 -o
credentials=/home/pi/.smbcredentials,uid=pi,gid=pi,iocharset=utf8,vers=3.0
```

> **Password with a `\`:** type it in `nano`, not with `echo` or `printf` – they change the backslash and mounting fails with `mount error(13): Permission denied`.

Permanently – add to `/etc/fstab` (a space in the share name is `\040`):

```
//192.168.1.10/Plex\040SSD1  /mnt/ssd1  cifs
credentials=/home/pi/.smbcredentials,uid=pi,gid=pi,iocharset=utf8,vers=3.0,_netdev,nofa
il  0  0
```

Check: `sudo mount -a && df -h /mnt/ssd1`. The wizard lists the share under "Network shares mounted on this Raspberry Pi".

One disk, two different paths

Each device sees the shared disk under its own path:

	Plex Pi	Downloader Pi
Mount point	`/mnt/ssd1`	`/mnt/ssd1` or e.g. `/mnt/plex_ssd`
"Concerts" folder	`/mnt/ssd1/Plex/Concerts`	`/mnt/plex_ssd/Concerts`

In Plex Downloader always enter the paths **of the device the panel runs on**. A path copied from the other Pi shows "Folder does not exist" – **Settings → Disks → Update paths** then moves it onto the right mount point (the same after importing settings from another device). Nothing changes in Plex.

One category, different folder names

A category is just a button in the panel – every disk has its own folder for it. E.g. the **Concerts** category: `Concerts` on one disk, `Live` on another. Enter the right path for each disk (or click the yellow **Create** next to an empty field), and in Plex add **both folders to the same library** (library → Edit → Add folders).

5. Everyday use

Downloading

Paste a direct link to a video file. The panel recognises the title, year, season and episode (you can correct everything), shows the TMDB card and the free space on the disk. Choose the disk and category and click **Download**. The file goes to a hidden temporary folder first and into the library in one move only when it is complete.

Advanced options: **edition** (`{edition-Director's Cut}`, up to 3 tags, needs Plex Pass), **file version** (`- 1080p`, several versions in one folder) and **TMDB ID** (`{tmdb-19995}`, exact match in Plex).

File manager

Two columns with posters. **Drag** a movie from one column to the other:

- same disk – instant move,
- another disk – copy → SHA-256 verification → only then the original is removed (**Skip SHA-256 verification** makes it about twice as fast and compares only the sizes).

TV shows can be moved whole, by season or by selected episodes. Change the order of the disks by dragging the tiles; the gear on a tile renames the disk.

Edit name and poster

The **pen** icon on a card (above the trash) opens the Edit window:

- **title, year, TMDB ID** (**Find in TMDB** button), **edition** and **files in the folder**,
- every file has a **type**: *Movie (version)* or a Plex extra – *Trailer, Behind the scenes, Deleted scene, Interview, Other extra*. An extra gets the name `Description-trailer.mp4` and Plex shows it under Extras, not as another version of the movie,
- file names change only after a field was edited – changing just the poster never renames anything,
- **Change** lists all changes and asks "Are you sure?".

Poster: *Own image…* (JPG, PNG, WebP), *Frame from the video* (6 frames to choose from; **Generate others** picks new ones; **Pick a time** shows frames around a time `h:mm:ss`, **+10 s** moves 10 seconds on) or *download the poster from TMDB*. Posters chosen by hand are kept when posters are refreshed.

Settings backups, export and import

- Every change of the settings creates a **restore point** (Settings → Backups), kept for 30 days.
- **Export settings** saves a JSON file (disks, categories, servers, options) – without the username, password and session keys; the TMDB key only on request.
- **Import settings…** shows what will change and lets you choose the sections. Disks from another device are unticked by default.

Activity log

Settings → Account → **LOG**: downloads, moves, deletions, renames. Clear and Export buttons.

---

6. Updating, repairing and diagnostics

Updating

```
cd ~/plex-downloader && ./update.sh
```

The header of the Settings window shows when a newer stable version is available.

Repair menu fix.sh

```
cd ~/plex-downloader && ./fix.sh
```

1. **Update** – installs a new stable version,
2. **Reinstall** – rebuilds the environment, settings stay,
3. **Clean reinstall** – factory settings (confirm with YES),
4. **Install / update BETA**,
5. **Leave BETA**,
6. **Server** – status, stop / start, restart,
7. **Additional modules** – Plex Media Server, Samba, SMB client, ffmpeg, Tailscale VPN (recommended), WireGuard VPN: status, install, update, stop until the next restart, restart, autostart on / off, uninstall,
8. **Install another version** – the stable versions since 1.2.5 with a short description; settings are kept (a copy goes to `.version-snapshots`), back to the newest: option 1 or `./update.sh`,
9. **Uninstall**,
10. **Cancel**.

Your movies and TV shows are never touched.

Raspberry Pi diagnostics

```
uptime                          # how long it has been running without a restart
vcgencmd get_throttled          # 0x0 = power supply and temperature OK
vcgencmd measure_temp           # temperature (trouble from about 80 °C)
free -h                         # memory
df -h                           # space on the card and disks
journalctl -u plex-downloader -n 50 --no-pager     # Plex Downloader log
dmesg -T | grep -iE "voltage|mmc|i/o error|cifs" | tail
```

To keep the log after a restart (useful when the Pi freezes):

```
sudo mkdir -p /var/log/journal && sudo systemctl restart systemd-journald
```

Then `journalctl -b -1` shows the entries from before the last restart. A `get_throttled` result other than `0x0` usually means the power supply is too weak.

---

7. Easier SSH login (keys)

On your computer (macOS, Linux, Windows 10/11 – Terminal or PowerShell):

```
ssh-keygen -t ed25519            # Enter, Enter (key passphrase optional)
ssh-copy-id pi@192.168.1.10      # you type the Pi password once
ssh-copy-id pi@192.168.1.20
```

On Windows instead of `ssh-copy-id`:

```
type $env:USERPROFILE\.ssh\id_ed25519.pub | ssh pi@192.168.1.10 "mkdir -p ~/.ssh && cat
>> ~/.ssh/authorized_keys"
```

Shortcuts – the file `~/.ssh/config` on your computer:

```
Host plex
    HostName 192.168.1.10
    User pi
Host zero
    HostName 192.168.1.20
    User pi
```

From now on `ssh plex` or `ssh zero` is enough. Stop a command with **Ctrl+C**, end the session with `exit`.

---

8. Access from outside your home (VPN)

The panel is meant for your home network. From outside – a phone on mobile data, work, a hotel – you connect through a **VPN**, a private network of your own devices. No ports are opened on the router and the panel stays invisible on the internet.

Tailscale is recommended: it connects your devices into a private network, has a free plan for home use, and you log in with a Google, Apple, Microsoft or GitHub account – always on the Tailscale website. The panel never sees or stores that login.

Installing Tailscale on the Raspberry Pi

1. Run `./fix.sh` → **7 Additional modules** → **Tailscale VPN** → **Install**. The script installs Tailscale from the official repository and allows the panel to log in to Tailscale itself.
2. At the end it asks whether to log in – you can skip that and log in from the panel (below).
3. If the installation stops with **404 Not Found** or **File has unexpected size** for system packages (e.g. `iptables`), a Raspbian mirror is at fault, not Tailscale. `fix.sh` then offers to download from the main Raspbian archive – answer **Yes**. By hand:

```
sudo cp /etc/apt/sources.list /etc/apt/sources.list.bak
sudo sed -i
's#http://raspbian.raspberrypi.org/raspbian#http://archive.raspbian.org/raspbian#'
/etc/apt/sources.list
sudo apt-get update && sudo apt-get install -y tailscale
```

If Tailscale was installed earlier in another way, choose **Update** in the same place – the panel then gets the permission to log in.

Logging in from the panel

1. At home open the panel → **Settings → VPN** → **Log in**.
2. A button with the link to the Tailscale login page appears, together with the same link as a **QR code** – log in on the computer or on the phone.
3. Choose the account and confirm adding the device. After a few seconds the tab changes to **“Connected to Tailscale”** by itself and the **VPN** badge in the header turns green.
4. Recommended: in the Tailscale console (login.tailscale.com/admin) → **Machines** → the Raspberry Pi's “…” menu → **Disable key expiry**. Otherwise Tailscale asks to log the Raspberry Pi in again about every 180 days.

Phone and computer

1. Install the **Tailscale** app (App Store, Google Play, tailscale.com/download) and log in to **the same account**.
2. In the VPN tab, section **“Open the panel over VPN”**, scan the QR code with the phone (Tailscale app switched on) or copy the address.
3. Add the panel to the phone's home screen like a regular website. The Raspberry Pi's VPN address (`100.x.y.z`) and its name (`raspberrypi.tail1234.ts.net`) are **permanent** – they do not change after a restart or when devices are switched off.

Away from home, switch the VPN on in the Tailscale app and open the saved address. At home you do not need the VPN – the panel works at its usual address, e.g. `http://192.168.1.10:8800`. Each address is a separate site for the browser, so you log in to the panel over VPN separately (the session lasts 30 days).

HTTPS in the VPN – a padlock instead of “Not secure”

Over VPN the connection is encrypted by Tailscale anyway, but the browser shows “Not secure” because the panel uses `http`. Tailscale can issue a real certificate for the panel:

1. **Settings → VPN** → **Switch HTTPS on**.
2. The first time, **Allow HTTPS in Tailscale** appears – open the link, allow HTTPS certificates for your account and click **Switch HTTPS on** again.
3. The panel is available at `https://raspberrypi.tail1234.ts.net` (no port number). This address is first in the list and in the QR code. **The first opening takes 10–20 seconds** while Tailscale gets the certificate; after that it opens at once.

Only your devices in Tailscale can open the HTTPS address. On the approval page Tailscale may also offer **Funnel** – the panel does not need it and never turns it on (Funnel publishes an address to the whole internet). If Funnel were on for the panel's address, the VPN tab shows a red warning with the command to switch it off. You can remove the Funnel permission in the Tailscale console: **Access controls → Definitions → Node attributes** → the row with the `funnel` attribute → “…” → **Delete**.

Where the panel is available

In the VPN tab you choose:

- **Home network + VPN** (default) – the panel opens at home and from your devices in Tailscale,
- **Home network only** – requests that come through the VPN (also the HTTPS address) are refused. Tailscale still works for other things (SSH, files, Plex).

Switching to "home network only" while you use the panel over VPN cuts you off at once – the panel asks first. You can switch back only from home.

What the VPN tab shows

- **Status:** connected / not logged in / switched off / service not running / not installed – with a hint what to do.
- **This device, VPN address and name, account.**
- **Tailscale version** with an **up to date** or **out of date** tag (and the current version number); a red **security update** tag means an urgent security fix. Update: `fix.sh` → 7 → Tailscale → **Update**.
- **Devices in your VPN** – which are online.
- **Log out / change account** – the Raspberry Pi leaves your Tailscale network; then **Log in** gives a new link and QR code (same or another account). Logged in again to the same account, the Raspberry Pi usually comes back with the same address; check in the console that key expiry is still disabled.

The **VPN** badge in the header: grey – VPN not set up, amber – needs attention, green – connected (with the VPN address). Clicking it opens the VPN tab.

Autostart and switching off

Tailscale starts with the system and connects without logging in again. You can change that:

- `sudo tailscale down` – disconnects the Raspberry Pi from the VPN (stays so after a restart); back: **Connect** in the panel or `sudo tailscale up`,
- `./fix.sh` → 7 → Tailscale → **Autostart: on → turn off** – the service does not start after a restart (turn it back on there).

WireGuard

> **Checked only in synthetic tests** – with simulated system commands, without a real WireGuard connection. If something works differently for you, please report it on GitHub (Issues).

WireGuard is for those who already run their own WireGuard server. Unlike Tailscale it needs **a port opened on the router** (usually UDP 51820) and keys and peers set up by hand.

- Installing the tools: `./fix.sh` → 7 → **WireGuard VPN** → **Install**.
- Configuration: the file `/etc/wireguard/wg0.conf` (keys, addresses, peers) – see the WireGuard documentation; start it with autostart: `sudo systemctl enable --now wg-quick@wg0`.

- The VPN tab shows a **WireGuard** section: active connections (e.g. `wg0`), their addresses and whether they start with the system. Connected peers and traffic: `sudo wg show`.
- Open the panel at the Raspberry Pi's address in the WireGuard network, e.g. `http://10.8.0.1:8800`. The **"Home network only"** setting applies only to Tailscale.

When something does not work

- **"The panel is not allowed..."** – run once on the Raspberry Pi the command shown in the tab (`sudo tailscale set --operator=<user>`) or `fix.sh` → 7 → Tailscale → **Update**.
- **"Service is not running"** – `sudo systemctl enable --now tailscaled`.
- **The phone does not open the address** – is the Tailscale app switched on and logged in to the same account? On the Raspberry Pi: `tailscale status` (device list; `direct` – direct connection, `relay` – through a Tailscale relay server: works, only slower).
- **The panel over VPN says "available only in the home network"** – at home switch to **Home network + VPN**.
- **The HTTPS address loads slowly the first time** – that is the certificate being fetched, 10–20 seconds.

---

9. Security in your home network

- **Do not forward port 8800** (or 32400 without a reason) on your router and do not put the Raspberry Pi in a DMZ – the panel is meant for your home network.
- For access from outside use a **VPN** (chapter 8): **Tailscale** (recommended, no open ports) or WireGuard.
- Use **strong passwords**: for the panel, for your Raspberry Pi account and for Samba. Better still, log in with SSH keys (chapter 7).
- In Samba use `valid users` – never share folders with guests.
- Keep the system up to date: `sudo apt update && sudo apt full-upgrade`, and Plex Downloader with `./update.sh`.
- The TMDB key and passwords are stored only locally (`config.json`, `~/.smbcredentials`) – never send these files to anyone.